

Reducing portfolio risk increases portfolio return. The 50 percent intermediate Treasury notes and 50 percent S&P 500 Index portfolio had an increase in return of 0.5 percent per year created by a reduction in return volatility of 2.9 percent.

CORRELATIONS ARE NOT CONSISTENT

Finding asset classes that have low correlation with each other is not easy. Financial articles and books frequently give tables or matrixes showing single historic correlation numbers between different asset classes in the matrix. Then the authors suggest using these static correlation numbers to make investment selections for your portfolio. In a sense, the authors are implying that the single historic correlation number will remain constant going forward. That is wrong. Correlations are dynamic, not static. They change over time.

It is very difficult to predict the direction any correlation will go in the future. Past correlations are not a reliable indicator of future correlations. The numbers can change frequently and without warning. Some asset classes may become more correlated with each other, and others become less correlated.

[Figure 3-9](#) is a visual representation of the shifting 36-month correlation between intermediate-term U.S. Treasury notes and the S&P 500 Index. If the correlation were fixed, the line would be straight across the chart. As you can clearly see, the line is hardly straight. There have been many periods of time when bond and stock returns moved in opposite directions from their averages (negative correlation) and several periods when they moved in the same direction with averages (positive correlation). There have also been many periods where there was no clear correlation between the two asset classes.

Over the past 20 years, the 60-month rolling correlation of returns between the five-year Treasury note and the S&P 500 has varied significantly. During the 1990s, the correlation was positive. Stock and bond returns moved in the same direction relative to their averages. Between 2000 and 2009, the correlation turned negative. The average correlation for the entire 20-year period was 0, suggesting that the two investments are not correlated.